

KRIPA SHANKAR

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EXPERIENCE

2025–2026 (Current): Coinbase, San Francisco, CA: Senior Director - Machine Learning & Eng - Head of Risk & AI/ML

Currently leading Risk AI/ML, Risk Platform, and Machine Learning Platform engineering and ML teams. Over the past year, modernized the risk and AI stack—slashing fraud losses by 87% YoY, virtually eliminating elder/romance scams (< 1/wk, > 90% recovery), scaling ML decisioning to 98%, and consolidating the organization into a single time zone. Key Highlights:

1. **Vulnerable User & Scam Protection:** Championed defenses for elders against predatory behavior (support scams, pig butchering, romance scams, and investment fraud), bringing monthly incidents down to < 1/week. Achieved > 90% recovery of stolen funds, successfully rehabilitating trust with global partners.
2. **P&L Impact & Model Precision:** Slashed user-loss ARR by 87% YoY and drove a record-breaking \$XXM drop in payment risk (steepest drop in 5 years). Reset Scam and ATO rates by an order of magnitude—from XX bps to 0.Y ppm—while scaling model precision by 3X.
3. **Customer Experience & Growth:** Optimized high-value user journeys by reducing friction by 57% and realized an 84% decrease in risk-related support volume.
4. **Platform & AI Stack Modernization:** Automated ML decisioning from 17% to 98% and integrated 12 new products in < 2 weeks. Launched user-facing **agentic chat defenses** and a **Financial AI Advisor**; pioneered deployment of Graph Foundation Models via Kumo.ai (influencing NVIDIA leadership in the lead-up to its acquisition).
5. **Engineering Velocity & Org Restructuring:** Embedded GenAI across developer workflows (84% of Risk/ML code is now AI-generated under 100% CI/CD), deployed internal agentic workflows (including a personal AI Chief of Staff and Caseflow AI agents with an alumnus's YC startup, now a top vendor), and orchestrated org-wide personnel restructuring to consolidate the team from three disparate time zones into a unified single time zone.

2022–2025: Google, Mountain View, CA: Director - Inventory Safety, TrustGraph, & AdSpam ML, Ads Privacy & Safety (APaS), Google Ads

Led teams with the business objectives below as the Eng & ML head of a suite of horizontal ML/AI products helping combat Ads Fraud for Mobile Ads, YouTube Ads, Search Ads, Display Ads, and DoubleClick:

1. **TrustGraph (TG) GraphMining AI:** Provided a technology platform to organize Google's anti-abuse and actor intelligence and make it easily accessible and useful to fraud-fighting teams. *Products: Learning Graphs, Node Inference, GNNs & Embeddings, Clustering, Dynamic Graph Mining, Model Understanding, TG Insights, TG Graph View, TG Data & Feature Store, TG Actor Knowledge.*
2. **GenAI - Behavioral AI for Invalid Traffic:** Invented LLM (Large Language Model) powered multimodal assessor (video, OCR, images, click-patterns) and Reinforcement Learning powered crawler to proactively combat behavioral abuse (\$XB impact) and reduce Mobile & Video UDR (UnDefended Rate) stemming from Incentivized-to-Click, Deceptive Traffic Acquisition, and Accidental Clicks, by 78%. *Products: BehaveAIVT.*
3. **GenAI - Trust & Safety Accelerator:** Accelerated T&S AI transformation with GenAI capabilities: *Products: Detecting Scams in Google Workspace with Few-Shot Text Classification, Better Advertiser Experience & Efficiency with Optimized Escalation Handling, Reverse Engineering Android Apps to detect Ad Fraud with Transfer Learning, Detecting Deep-Fakes on Google Search with PaLI/Gemini, AdSpam Consultant: Automate 80% of responses to appeals & consults from advertisers at 95%+ accuracy with automatic comprehension.*
4. **AdSpam ML Platform:** Accelerated the end-to-end ML development lifecycle to mitigate attacks and force-multiply the impact of regulatory-compliant, privacy-compliant, secure AdSpam defense workflows. *Products: Multiverse, ML360, Auto-retraining, Model Explainer, Signal Warehouse.*
5. **AdSpam Emerging ML (EML):** Invented new ML models in response to emerging threats and emerging methods, and collaborated with AdSpam/APaS teams to productionize, launch, and support them. *Products: pModels, Hybrid Models, Termination Labels based supervised models, Transfer learning Teacher Model as a service, YouTube Representation Learning Model Service.*
6. **AdSpam Measurements (ASM):** Accurately and comprehensively measured business-critical metrics reflecting AdSpam's defense quality (incl. defense coverage, defense gap, defense friction, customer pain). *Products: UDR Risk Model, Auto Validation Ensemble Model Service*

Key Wins Delivered:

- **ARG + Cloud:** As the co-lead for the ARG (Ads + Research + GoogleDeepMind) partnership program, drove 18 joint goals to completion in 2024 and landed the strategy for 2025–2028. Externalized core graph innovations from TG via Google Cloud as Spanner Graph. As the Area Tech Lead for ML & LLMs for AdSpam & TrustGraph, drove 2 hackathons, 1 newsletter, and MBRs.
- **TrustGraph:** Redefined the flywheel and launched 12 ML product features and teams to deliver \$800M in Anti-Abuse-\$ Saved (+65% over target, ↑55% YoY, ↑180% since taking ownership) and delivered \$2.5B in lifetime impact (3X-ing the

same in 2 years). Amplified impact across 40+ teams / 5 PAs across Google. Drove MAUs up 159% and CSAT from 3.6 to 4.1.

- **AdSpam ML:** Owned the VP+ anti-abuse goal to deliver \$875M impact by EOY24 (using AI for Actor, Behavioral and Foundational Model innovations) and landed 2X impact of \$1.9B in Oct '24 (2.5 months ahead). Increased multi-modal LLM based crawling coverage for Ads in Apps+Mobile from 23% to 72%, and incrementally captured \$550M (+375%) in abuse.
- **Regulatory & Privacy:** Achieved 100% DMA (Digital Markets Act) and Privacy compliance (21 systems, 727 Defense Pipelines, XK+ signals), 5.5 months ahead of schedule by partnering with legal and 5+ core orgs in Google. Delivered 142% replacement effectiveness against a target of 70% for replacing privacy-sensitive signals (ML Features).
- **Trust & Safety:** Launched AdSpam Consultant and expanded the same as an AI rater/agent to reduce level-of-effort by 57% and increase throughput by 65%. Increased coverage of ML-based measurement of Undefended Rate by inventing a continuous monitoring solution to cover \$XB+ blind-spot networks (revenue-wise) and expanded the same to become the Quality Monitoring solution (SyGT) for all defense pipelines too.
- Baselined and increased ESAT (Employee SAT) from 7.7 to 9.2; landed 2 L8 Director level promos (Eng & PM) and 7 L6+ promos (17 total).
- Brought together 150+ participants across 5 cities for an APaS Summer Olympics event.
- Speaker, sponsor, and dinner host for 3 AI-for-everyone events to bring together a community of Bay Area AI talent.
- **Ads Allyship Award** Nominee for sponsoring and running the APaS LLM for Women working group.

2009–2022: **Amazon.com**, Seattle, WA

2021–2022: **General Manager (Engg, Product & Applied Science), Secure AI Foundations, Alexa AI**

From Jun '21 to Sep '22, led a BizTech org of 163 including an Applied Science team of 23, Engineering team of 95, Product/Program Management team of 22, and a Support Engineering team of 10 (with 2 L7 Sr. SDMs, 1 L7 Sr. Mgr of Product, 2 Principals, 5 SDMs, 4 Sr. SDEs) building Intelligence Augmentation using AI for Alexa AI.

Highlights:

- **Intelligence Augmentation:** Improved Alexa's Intelligence by 700 bps to 13% across 14 intelligence success metrics by augmenting labeled data and launching data-generation solutions.
- **Automation:** Reduced costs for data-services by \$15MM YoY by launching 23 ML-in-the-loop (MLITL) automation models and systems WW. Single-threaded owner for MLITL programs for ASR (Automated Speech Recognition) and NU (Natural Language Understanding).
- **Privacy:** Improved the privacy posture for Alexa by launching distorted-voice-recording systems and voice-conversion models for secure sites and owned the SVP goals for privacy, intelligence and automation.
- **Science:** Delivered a paper that was accepted at the Association for Computational Linguistics (ACL) for contrastive meta-learning method to estimate human label confidence scores and reduce data collection cost.

2019–2021: **Senior Manager (Engg & Applied Science), ML Center & Alexa Lab, Alexa AI**

From Nov '19 - Jun '21, led 2 orgs of 61 (incl. 2 L7 Sr. SDMs, 5 SDMs, 7 Sr. SDEs, 2 Research Scientists, 3 TPMs, PMT, 56 SDEs, 8 PS) building ML Center a.k.a. SensAI MLX (ML experience that brings together 97 Core ML Services) and Alexa Lab (the online experimentation platform for Alexa & Devices) for Alexa AI.

Highlights:

- **Sensai Console & Search:** Launched Sensai Console Platform that can provide a Single-Pane-of-Glass (SPOG) UX for all Alexa ML services, APIs & tools. Delivered Sensai Search to help discover all ML artifacts (models, datasets, workflows, compute containers/snapshots) with lineage. Grew the MAUs from zero to 2,424.
- **Sensai IAM: 1-Day New Modeler Onboarding:** Reduced new Alexa modelers' onboarding time from 13 days to 1 day; enabling modelers to contribute from day-1 (SVP Goal). Launched SPOG Sensai IAM Console for managing access (3400 MAUs) for 135 Alexa & Devices teams. Reduced time taken to register/access (p90) critical (security classification) ML artifacts from 20 days to zero (instantaneous).
- **Sensai Visual ML:** Launched **SageMaker Notebooks** for ML development, and HoverDesktop as the single window to the secure ML compute environment with 40+ features, 1322 MAUs (+707% QoQ), and reduced launch steps from 11 to 1.
- **Alexa Lab:** Grew customer teams to 76 (+51% YoY), experiments to 2,437/year (+122% YoY), metrics to 608 (+110% YoY), Experimentation Rate to 24.5% (+47% YoY), and Bar Raisers from 4 to 23. Reduced Scorecard generation time (p90) from 124 hours to 30 hours (-75% YoY), and launched: (i) Alexa Lab APIs as an external product to 1P & Alexa Skill Developers via Alexa Skills Kit, (ii) Universal Guardrail Procedure (UGP) methodology to guide business decisions, and (iii) Reflex, a No-Code o-SDE effort solution to setup and run A/B experiments of models/CX.

2017–2019: **Senior Manager, Identity Risk & Core ML, Customer Trust & Partner Services (CTPS)**

2014–2017: **Manager III, Fraud Prevention & Identity Risk, Risk ML**

2009–2014: **Applied Scientist III < Applied Scientist II < SDE, Risk ML & Customer Account Protection**

Led 2 orgs of 44 SDEs & MLEs (2 Principal Engineers, 5 SDE-3s), 5 SDMs, & 2 TPMs across: (i) Core ML Services, (ii) Fraud Prevention, and (iii) Identity Risk Engineering teams in Transaction Risk Management Systems (TRMS) & CTPS.

Highlights:

- Authored, achieved funding and completed successful 3-Year-Plans (3YP) from ideation to launch such as:
 - **NoPassword**: Reduced customer auths that ask for password by 9B/month (↓87%) without impacting fraud rates by leveraging customer's **behavioral genomes**, **behavioral fingerprints**, and **fraud-graphs**, enabling frictionless shopping experiences with higher assurances and lower fraud risks.
 - **Amazon Smart Armor Graph & On-The-Fly ML Feature Engineering Platform**: Initiatives to leverage the digital ecosystem of devices that surround and define a customer to help protect the customer automatically, and prevent abuse risks.
- Delivered: (i) End-to-End **Secure AI Sandbox** for ML Scientists (MLS) to experiment securely without putting customer data at risk, (ii) **Unified ML Catalog** with goals to increase reusability of 50K+ ML Features using data-lineage, (iii) **Intelligent ML Pipelines** for MLs to auto-deploy ML solutions to production with the right safety nets, and (iv) **Automated Model Building** using ML-as-Code building solution with the goal of automating 100% of all model refreshes and retraining tasks to meet the Payment Fraud Risk SVP-level goals for 2018–2019.
- Owned the Amazon Retail S-Team (Senior VP Leadership Team) goal to reduce fraud rates across **buyers, sellers, vendors, developers & AWS customers** stemming from Account Takeovers (ATOs) & credit-card fraud and helped reduce the same by 70% over the course of 2014 to 2018. Met and **beat the S-team goal for 3 years in a row**.
- Owned authoring & driving **Quarterly Business Reviews (QBRs) and a Cross-Functional Steering committee** spanning VPs & Directors across Customer Service, Identity Service, InfoSec, Reviews / Community Shopping, Website Platform from 2016-2018.
- Certified **Amazon Security Certifier** with 5+ years of certification experience and built pools of certifiers.
- **Hire & Develop the Best**:
 - Achieved the **highest Tech Survey results** scores across CTPS's 110+ Sr. Mgrs in an org that is 16K people strong, with a score that was 17% above Amazon's benchmarks.
 - Grew the Fraud Prevention Identity Risk team from a team of **4 in 2014 to 37 in 2018**, and the Core ML Services team (for Fraud Prevention) from a team of **19 in 2017 to 36 in 2019**.
 - Established the SAIF MLX (Sensai + Alexa Lab) org and grew it from **12 in 2019 to 61 in '21**.
 - **Bar Raiser** with experience from **~980 interviews** with mentoring and graduating Bar-Raisers-in-Training.
 - Employee growth/promotions driven: **1 L7 SDM, 2 L7 Principal Engineers, 1 L7 PM-T, 10 L6 SDE-3s, 3 L6 TPMs, 3 L6 SDMs, 16+ L5 SDE2s**. Currently working on 2 PE and 1 Principal PM-T promos.
 - Bootstrapped a new Amazon remote office location in **San Diego**. Helped build & grow the Identity Risk Fraud Prevention **Data Science** Team, and Bot Detection Engg Team by transitioning services from SEA to SAN.
- **AWS Launches**:
 - **re:Invent 2017 & GA 2018**: Launched **AWS Cognito Frontline** with adaptive (risk-based) authentication. Expanded risk ML services (drove SVP approval) from Amazon retail to provide "Protect Customer Account Pools" (named Cognito Frontline) as an AWS service in partnership. Grew the same to 7B+ customer accounts (by launching for Intuit, LucidChart, Washington Post, Pinterest, & Clyp.it).
 - **re:Invent 2019**: Launched **Amazon Fraud Detector**, a fully managed service that makes it easy (automatically with just a few clicks and no prior ML experience) to identify fraudulent online activities such as online payment fraud and fake accounts faster.
- Honorary & Program Committee member of **eFraud Global Forum (eFG)**, 16 years running: eFG is a closed door, invitation-only forum of leaders (CISO, Fraud Preventions Heads, VPs) across all industries that encourages learning through sharing information about how online fraud occurs and how to prevent it using best practice examples from leading private & government organizations.
- Forged partnerships with organizations such as: (i) OpenId's Risk and Incident Sharing and Coordination (**RISC**) WG to combat fraud and identity risks together with Google, Microsoft & other identity providers; and (ii) Government orgs such as National Cyber-Forensics & Training Alliance (**NCFTA**) to bring in intelligence data.
- **Patent holder & author** for ideas such as: Automatically Adding Backup Verification Devices to Accounts, introducing a cool-down period to Password reset workflows, and Proof of Work Challenges to drain malicious bots of their CPU & GPU resources using Bitcoin-mining/Hashcash-based challenges.
- Launched an Amazon-wide Robot Detection & Mitigation program that helped shed unnecessary robot traffic and saved Amazon **\$54MM** in hardware costs in 2016. Innovated low-cost queuing solutions (for 1/35 times the cost of equivalent AWS solutions) that could process **1.2MM TPS** of real-time clickstream traffic using a custom-built Apache Kafka & Cassandra architecture.
- Led a team of 20 SDEs in BLR and migrated the team from BLR to SEA in 2015 while building systems (intelligent Know-Your-Customer & Sanctions Screening systems) to mitigate **money-laundering risks**.
- Other notable launches include:
 - Amazon **CAPTCHA Service**, capable of presenting CAPTCHAs of varying toughness levels on-the-fly (audio + animated CAPTCHAs), with built-in behavior track and fingerprinting.
 - New **Password Reset UX** that reduced customer contact rate by 60% YoY.
 - **Mobile & Device Fingerprinting** to profile mobile traffic better and reduced ATO rates YoY by 12%.
 - **Transaction Intent Verification** as a new method for risk & ML based user authentication that not only provides

assurance that the user was involved with a transaction, but also that the transaction was what the user intended.

- **Customer Challenges** like Risk-based One-Time Passcode challenges & Email Verification challenges to mitigate fraud risks and **Remote Logout**, as a capability for customers to manage their devices.
- **Amazon Authentication Standard, Device Authentication Spec** and a compliance program for the same.
- **Fraud Cluster Detection, Phishing Detection, Account Compromise Detection & Bot Detection Systems**: Adapted and built different ML models (Naive Bayesian - 2009, Logistic Regression, Random Forest, Decision Trees, DNN, CNN, PMML, LSTM, GNN - 2019) expanded fraud capture rate by 45%-70% across 34 Amazon businesses. Engineered systems that could auto-upgrade core models with no human intervention. Solved clustering problems (**Repeat Offenders**) for detecting malicious collectives.
- Achieved **highest leadership ratings of “Top Tier”, “Role Model” & “Outstanding”** across L7, L6 and L5 levels in Amazon; and have been certified via Tech Leadership Accelerator Programs & LeAP trainings.

Google Inc., Mountain View, CA

2008–2009: **Software Engineering Intern & Co-Op**

- Studied and helped develop a service to track and find security vulnerabilities in Google Docs, Google Gears.
- Conceptualized & developed Tellurium, a lightweight proxy server that manages & simulates network traffic using ML algorithms to help execute penetration attacks on web-applications (specifically Google Docs).
- Extended the tool to mock collaborative scenarios (without using multiple browsers) and support offline analysis.

PURDUE UNIVERSITY, West Lafayette, IN

2007–2009: **Research Assistant, Internet Systems Lab** (Advisor: Prof. Sanjay Rao)

- Research Areas: P2P Networks, DHTs, NATs, Network/Web Security

2007–2009: **Research Assistant, Rosen Center for Advanced Computing**

- Involved in the design of Cyber-infrastructure for End-to-End Environment Exploration (C4E4, Grid Computing)
- Programmed a distributed service oriented workflow integration framework over Purdue’s TeraGrid data management system

2007: **Web-Programmer, Department of Biochemistry**

- Designed and re-built more than 200 pages using a custom code-generator.
- Added support for web based transactions. Programmed a configurable XML rule-based accessibility checking tool.

SAP LABS, Bangalore, India

● 2005–2007: **Development Specialist, Emerging Solutions**

- Recognized as “Top Talent 2006-07”—a title given to top 2% of global SAP employees
- Received “Performance par Excellence” employee recognition award; achieved a performance rating of 123%
- Collaborated regularly with teams from Israel, Germany and Palo Alto (USA) and show-cased strong API designs
- Pioneered a software development kit that brought down development time by 60%
- Key framework developer in Duet (a joint venture with Microsoft), Groupware/Mobile Integration & NetWeaver projects
- Engineered security enhancements and innovative XML binding solutions that boosted system performance by 34%
- Participated in many leadership skills training programs; was on an accelerated architect growth track.

EDUCATION

PURDUE UNIVERSITY, West Lafayette, IN

2007–2009: **Master of Science - Computer Engineering**

- Specialized in areas of Networks & Computer/Information Security & Machine Learning
- CERIAS (Center for Education & Research in Information Assurance & Security) Member
- Bharat Petroleum Scholar; awarded BPCL Scholarship for pursuing graduate studies

UNIVERSITY OF MADRAS, Chennai, India

2000–2004: **Bachelor of Engineering, Electronics and Communications**

- Ranked second in a class of 140. Top 1% of the university. First Class with Distinction
- Chairman - IEEE Student Branch, Madras Chapter
- General Secretary - Electronics & Communications Department (highest office-bearer post)
- Member of Usha Martin Academy of Communication Technology (IIT Madras) & IEEE